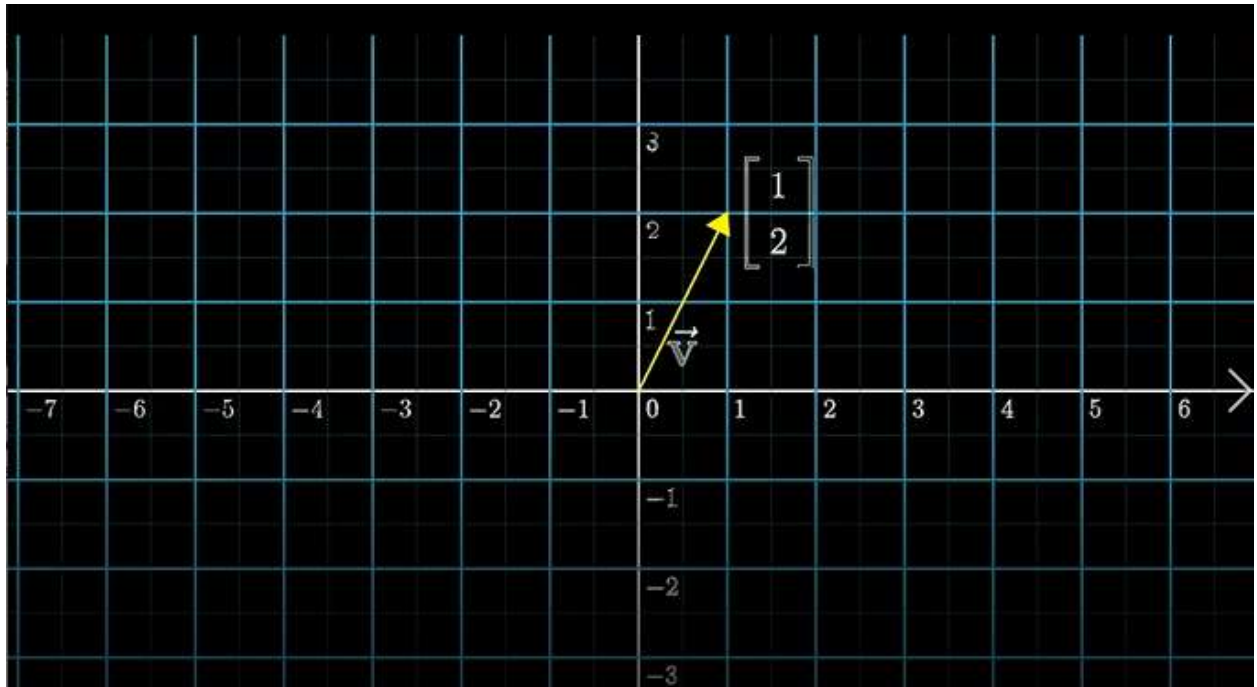


VECTOR

THE FUNDAMENTAL OF ANY LINEAR ALGEBRA IS VECTOR

What is the vector?

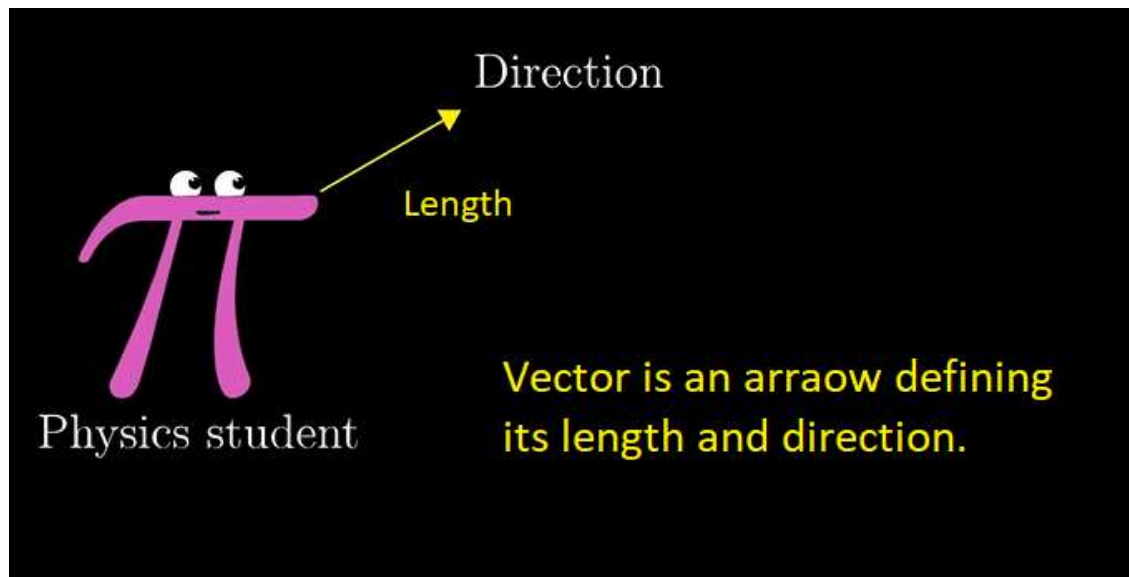


v is a vector having co-ordinate $(1,2)$

There are Three ideas of a Vector

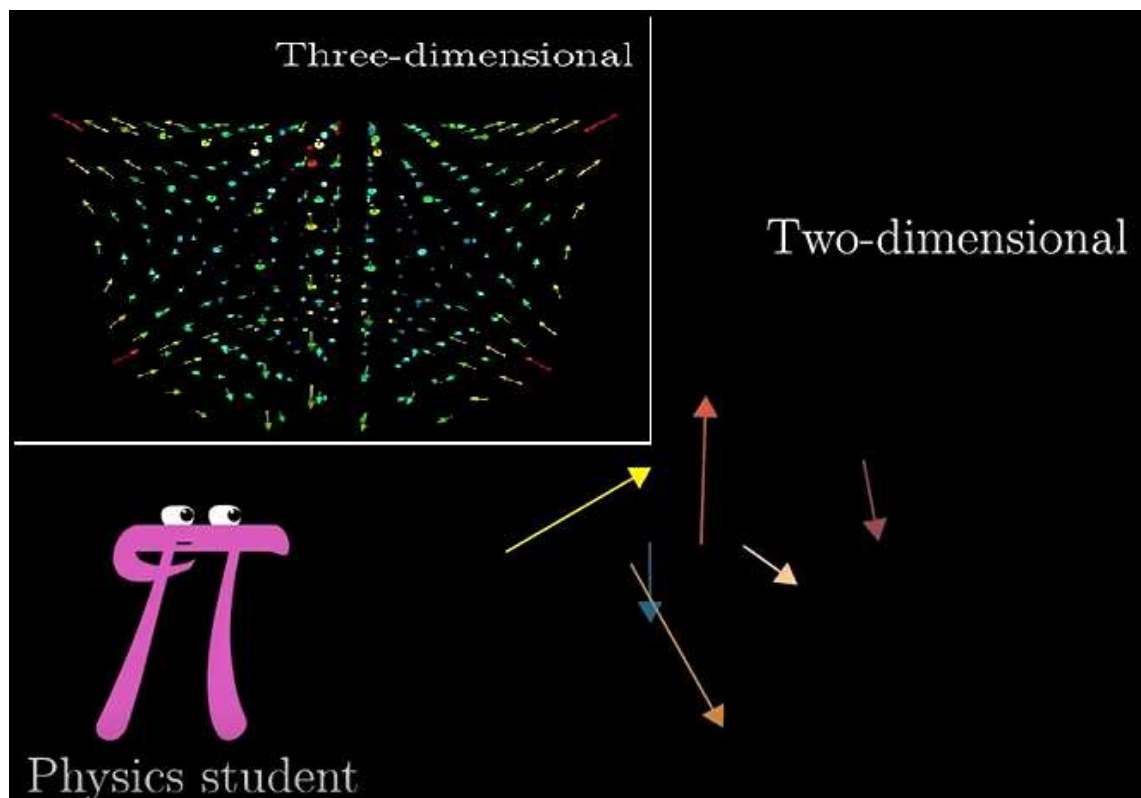
- (1) Physics prospective
- (2) Computer Science Prospective
- (3) Mathematician Prospective

PHYSICS STUDENTS



You can move this vector all around and it still the same vector.

Vector Can be Two dimensional and Three dimensional also.



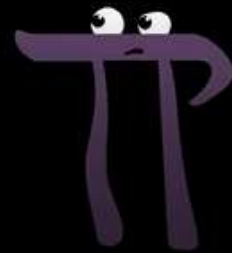
COMPUTER SCIENCE PROSPECTIVE:-

VECTOR IS A LIST OF NUMBERS.

It is a type of metrics.

Vectors $\Leftrightarrow$ lists of numbers

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 5 \\ 0 \\ 0 \\ -3 \end{bmatrix} \quad \begin{bmatrix} 2.3 \\ -7.1 \\ 0.1 \end{bmatrix}$$



CS student

Vectors $\Leftrightarrow$ lists of numbers



Square footage: 2,600 ft²
Price: \$300,000



$$\begin{bmatrix} 2,600 \text{ ft}^2 \\ \$300,000 \end{bmatrix}$$

Here the price of the house is
depends on the size of the
house

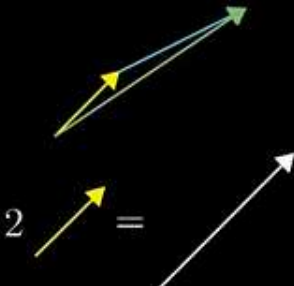
But we can not reverse the
order of the Matrix

$$\begin{bmatrix} 2,600 \text{ ft}^2 \\ \$300,000 \end{bmatrix} \neq \begin{bmatrix} 300,000 \text{ ft}^2 \\ \$2,600 \end{bmatrix}$$

From Mathematics Point of View:-

We can do multiplication and addition with Vectors(Metrics).

From Mathematics point of view we can add the vector , multiply the vector


$$\vec{v} + \vec{w}$$
$$2 \vec{v}$$
$$\begin{bmatrix} 3 \\ -5 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3+2 \\ -5+1 \end{bmatrix}$$
$$2 \begin{bmatrix} 3 \\ -5 \end{bmatrix} = \begin{bmatrix} 2(3) \\ 2(-5) \end{bmatrix}$$


Mathematician

The Multiplication of Vector and Addition of Vector Play very important role in Algebra.

Product of two vectors: -

Product of two number = Number.

Product of Two Matrix = Matrix

Multiplication of Two Vectors:-

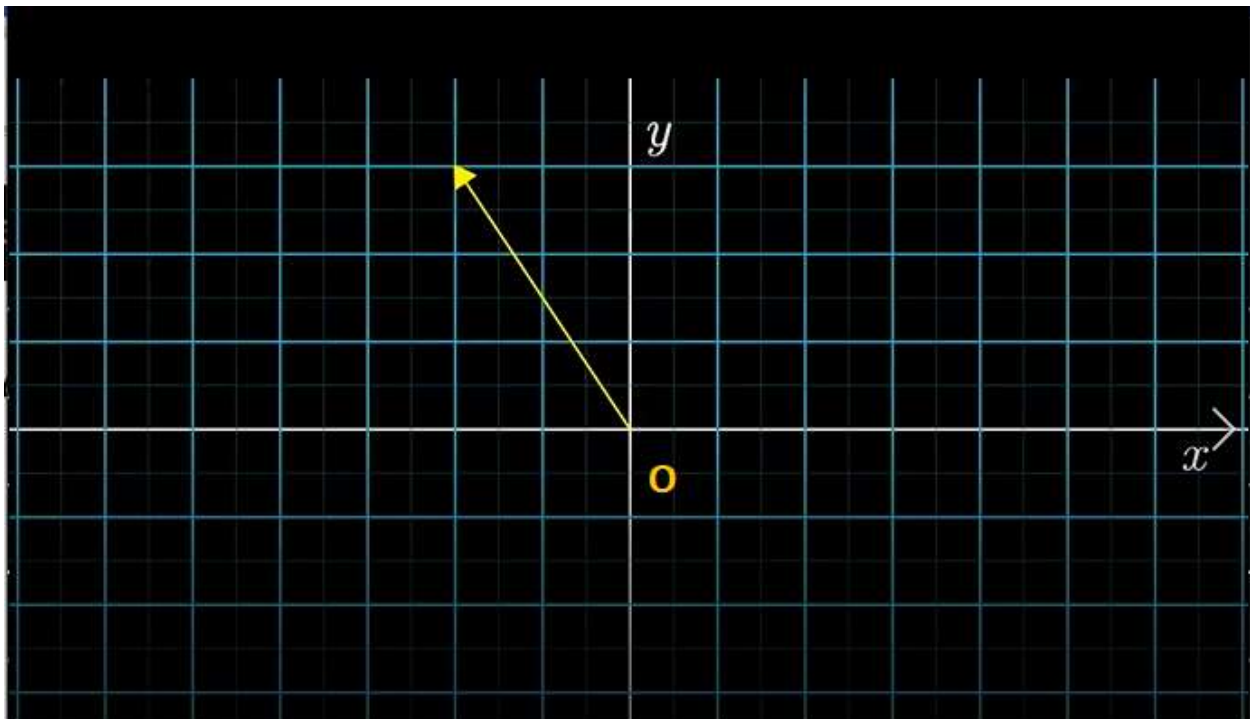
(1) Scaler or Dot product $\rightarrow$ scaler

(2) Vector or Cross Products $\rightarrow$ vector

Scaler or Dot product: -

Vector or cross product: -

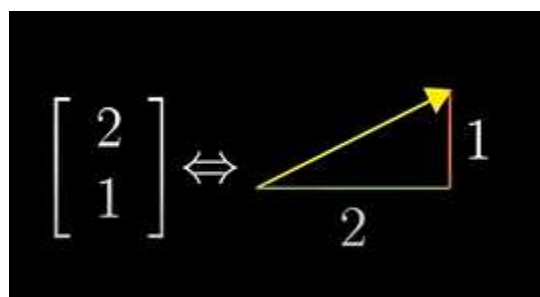
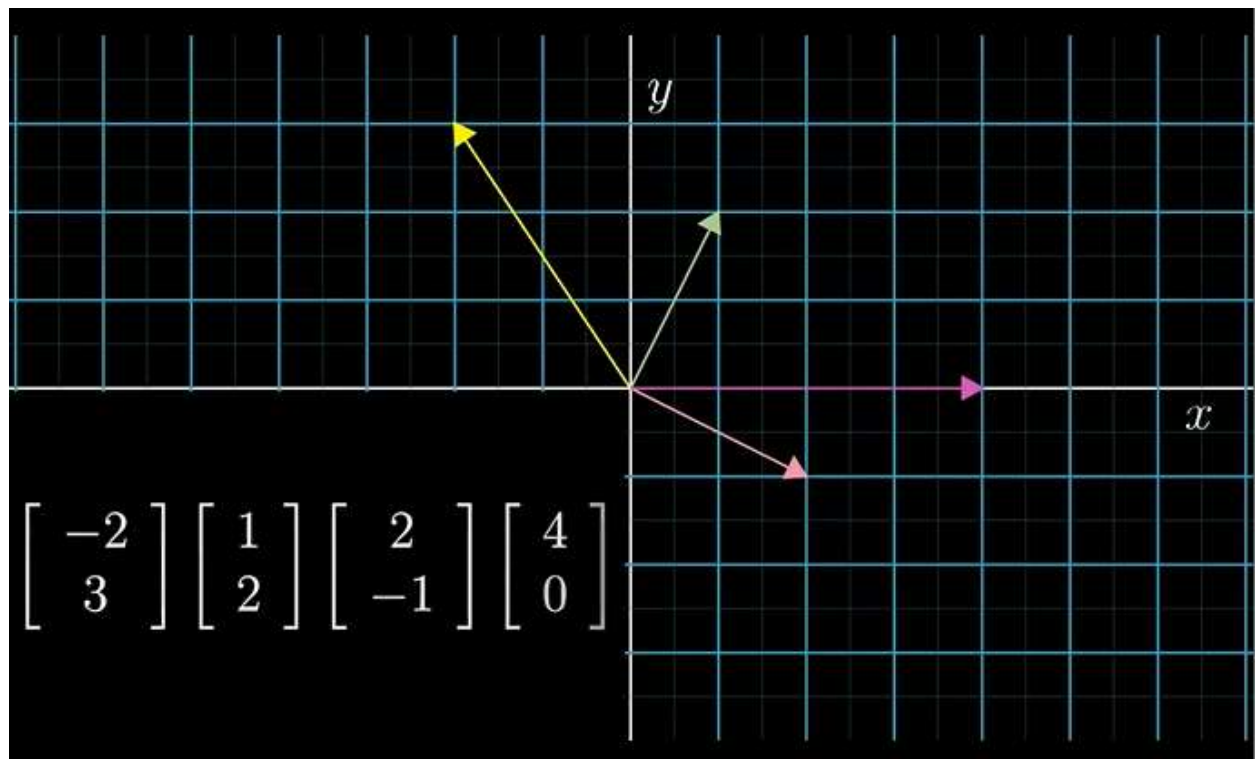
- In Physics Vector can be anywhere in the space not necessary the Tail is at the Origin O.
- But in Linear Algebra the Tail of the Vector is considered always at the origin.

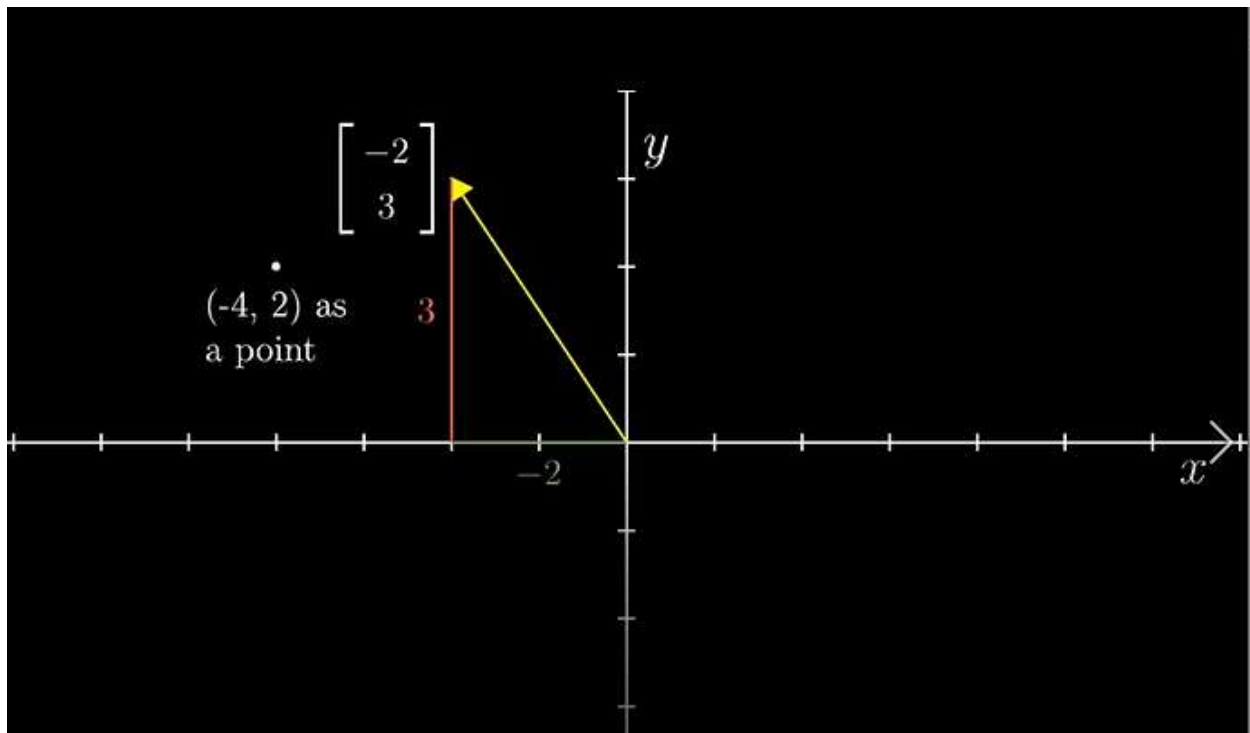


Consider this arrow inside a co-ordinate system in X,Y plane
tail in the Origin

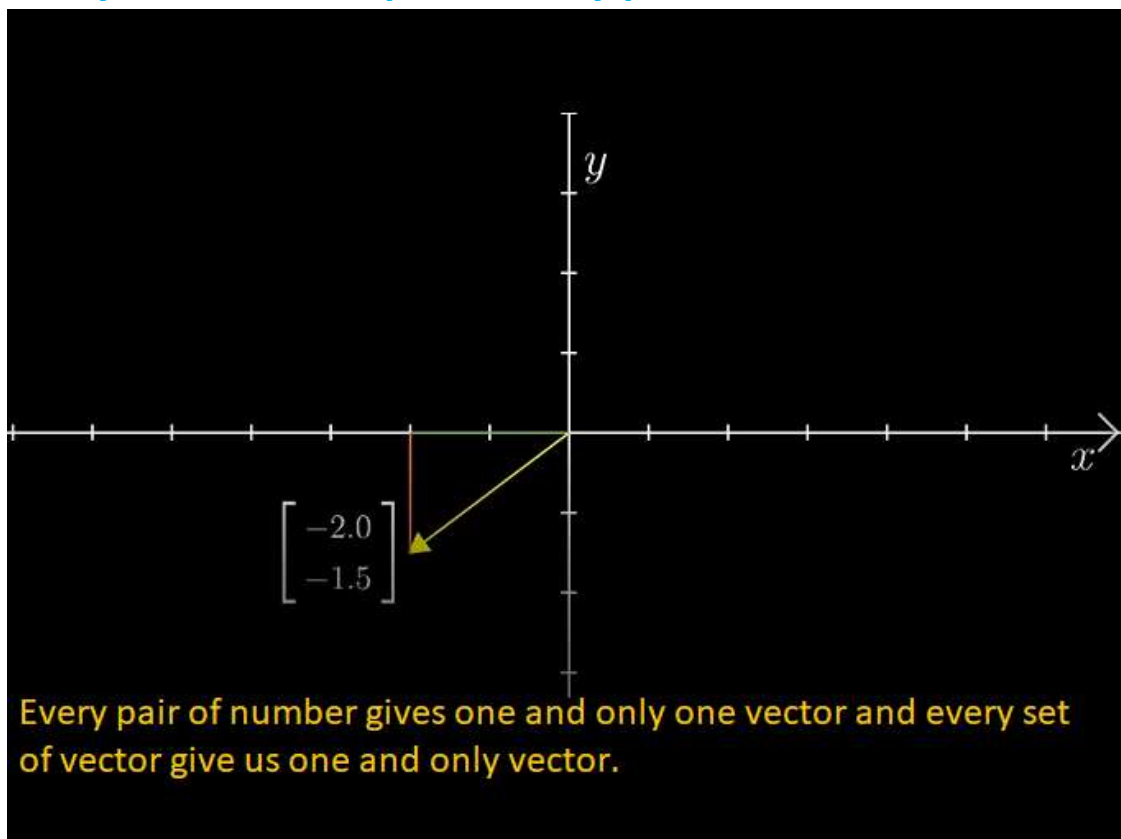
We can consider the direction of vectors as the co-ordinate of the system.

Understanding Vector Co-ordinate system

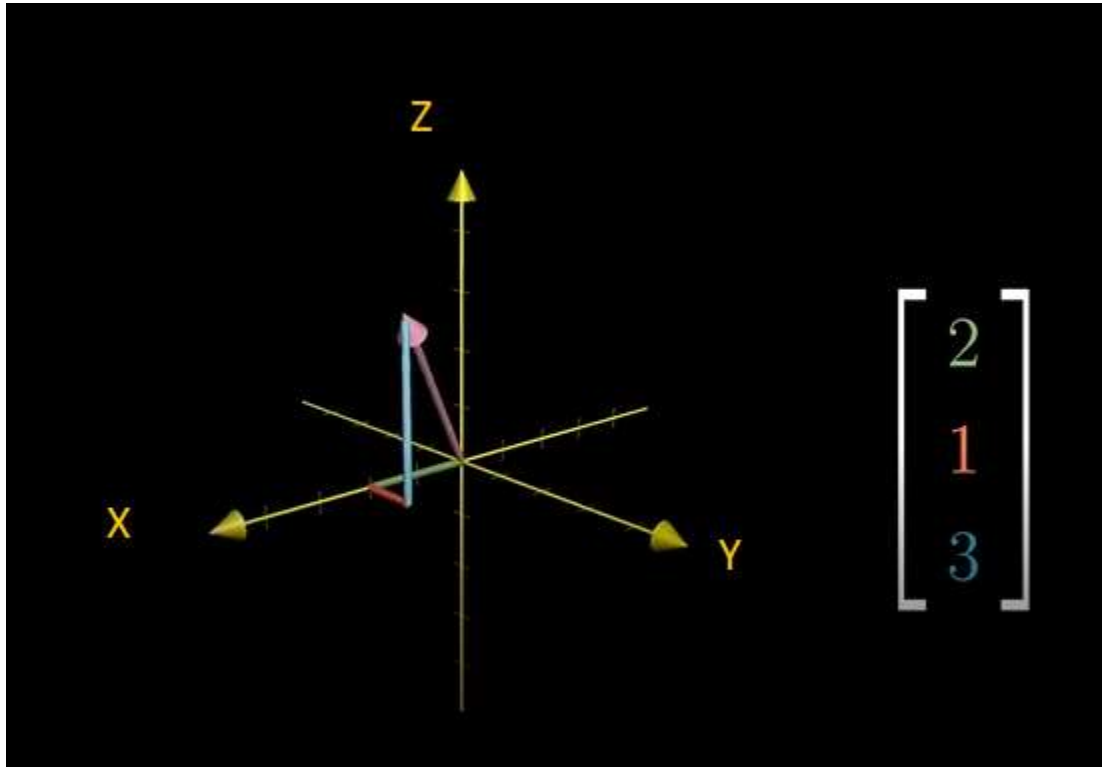




Every vector is unique & every pair of number is also unique.

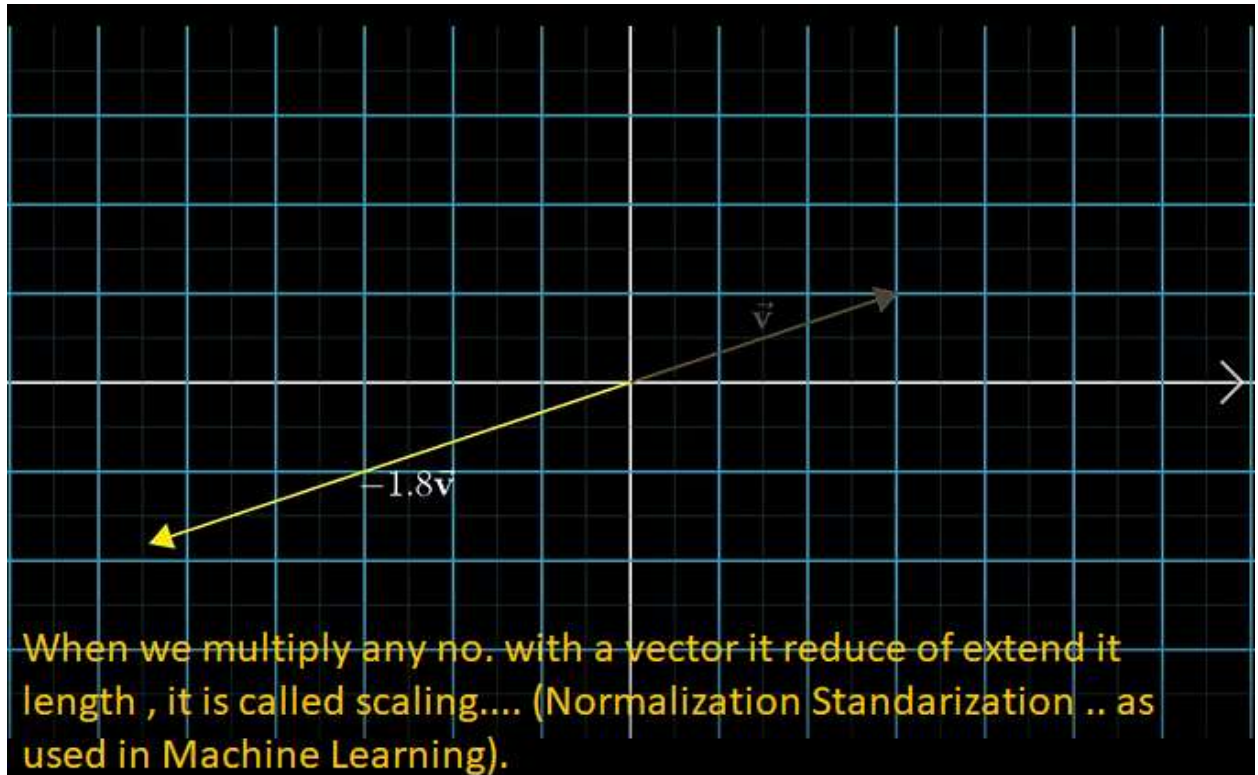


Three Dimension Plane

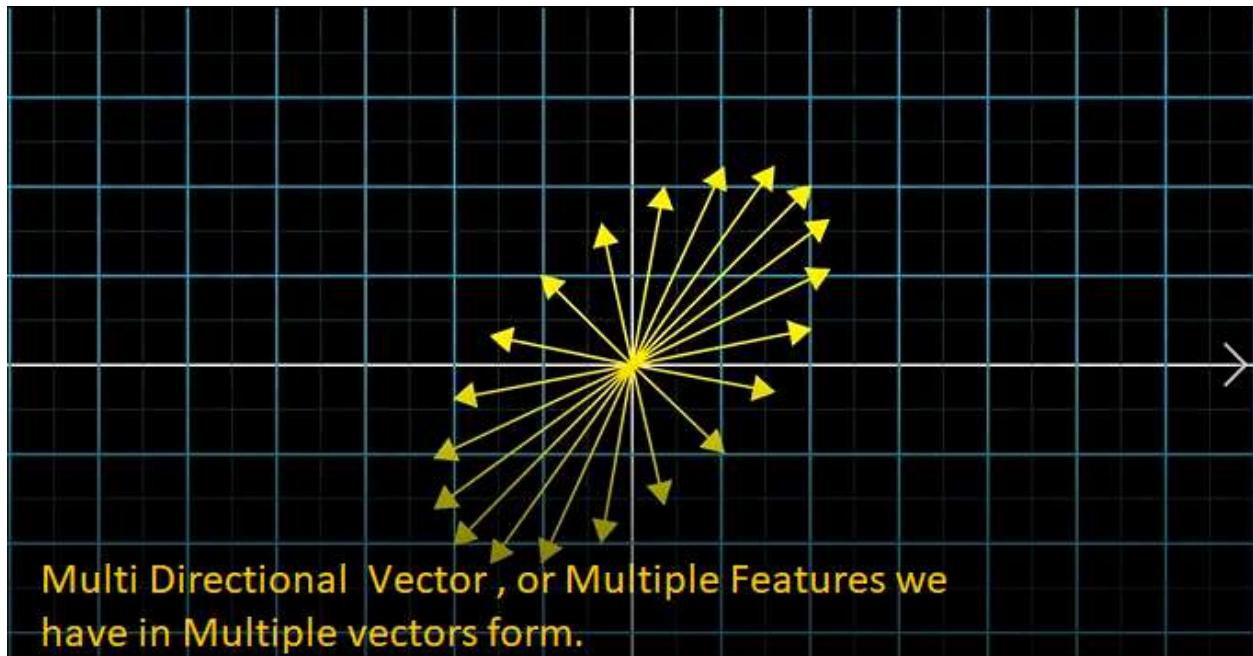


Here $X=2, Y=1, Z=3$.

Scaling a Vector



DATA SCIENCE POINT OF VIEW multiple vectors /metrix

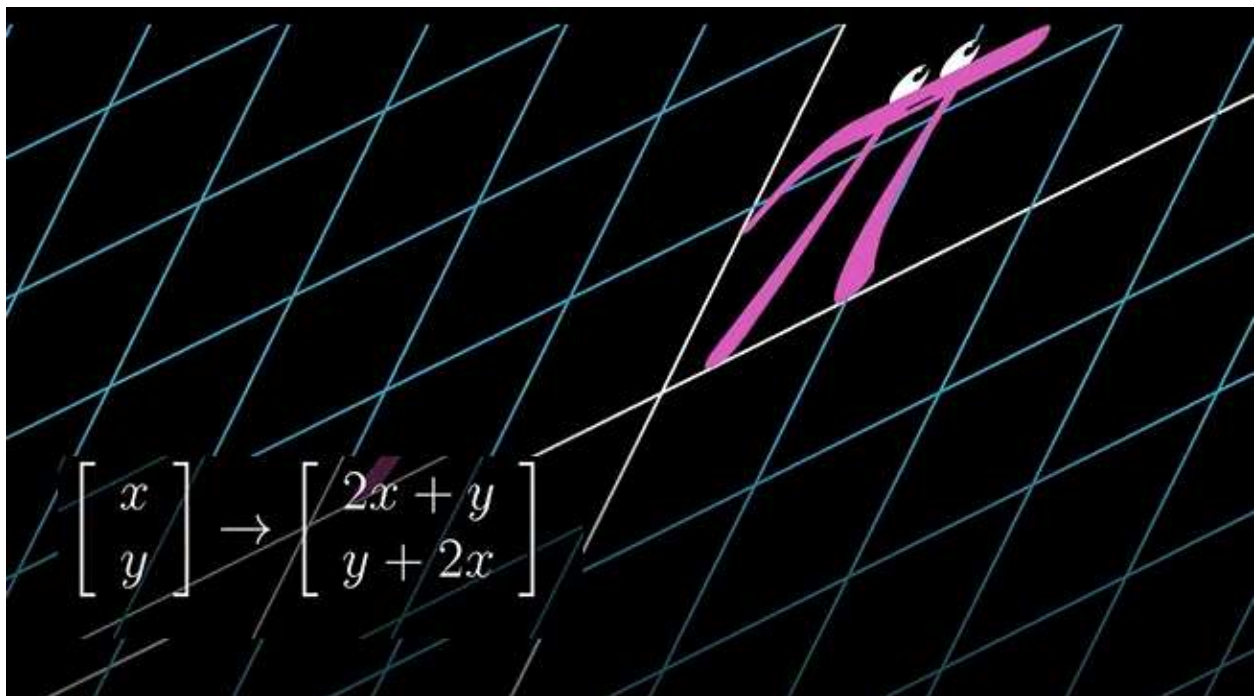
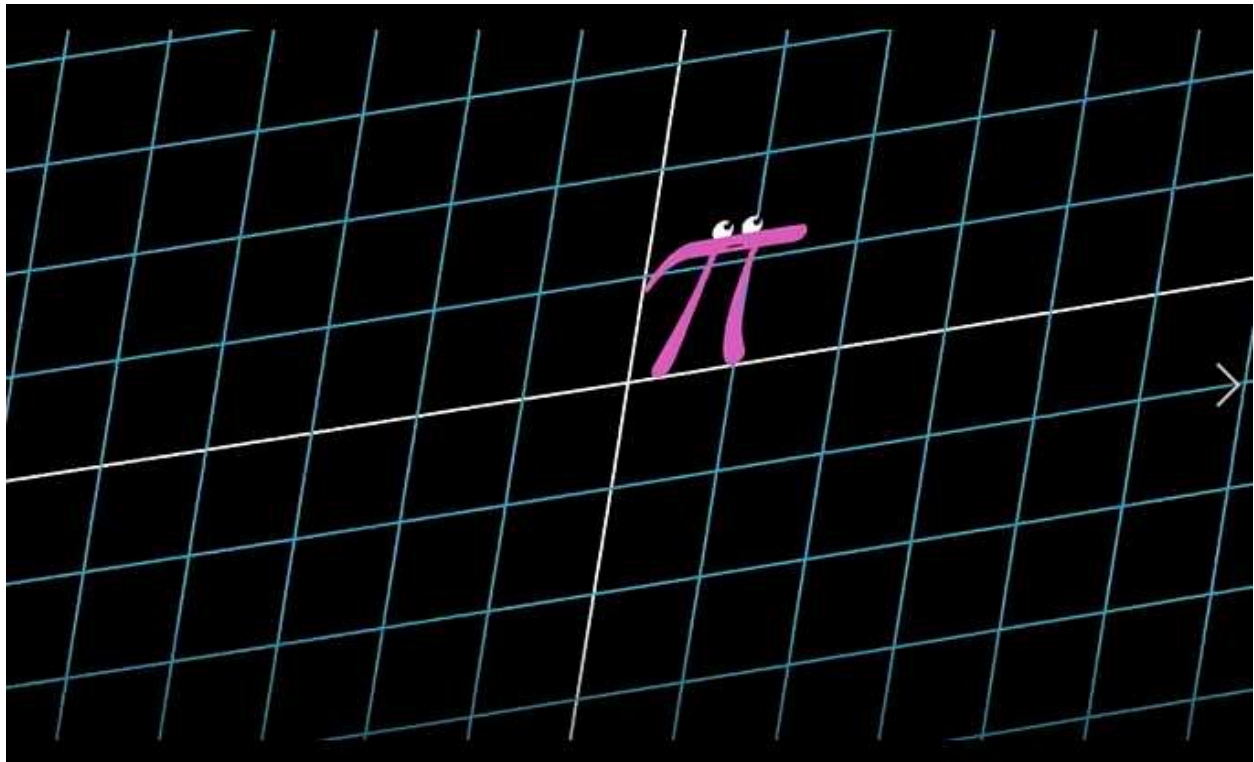


$\begin{bmatrix} 2.00 \\ 2.00 \end{bmatrix}$	$\begin{bmatrix} 1.59 \\ 2.21 \end{bmatrix}$	$\begin{bmatrix} 1.03 \\ 2.21 \end{bmatrix}$	$\begin{bmatrix} 0.36 \\ 1.98 \end{bmatrix}$	$\begin{bmatrix} -0.34 \\ 1.56 \end{bmatrix}$
$\begin{bmatrix} -1.01 \\ 0.99 \end{bmatrix}$	$\begin{bmatrix} -1.58 \\ 0.32 \end{bmatrix}$	$\begin{bmatrix} -1.99 \\ -0.38 \end{bmatrix}$	$\begin{bmatrix} -2.21 \\ -1.04 \end{bmatrix}$	$\begin{bmatrix} -2.21 \\ -1.60 \end{bmatrix}$
$\begin{bmatrix} -1.99 \\ -2.01 \end{bmatrix}$	$\begin{bmatrix} -1.58 \\ -2.21 \end{bmatrix}$	$\begin{bmatrix} -1.01 \\ -2.20 \end{bmatrix}$	$\begin{bmatrix} -0.34 \\ -1.97 \end{bmatrix}$	$\begin{bmatrix} 0.36 \\ -1.55 \end{bmatrix}$
$\begin{bmatrix} 1.03 \\ -0.97 \end{bmatrix}$	$\begin{bmatrix} 1.59 \\ -0.30 \end{bmatrix}$	$\begin{bmatrix} 2.00 \\ 0.40 \end{bmatrix}$	$\begin{bmatrix} 2.21 \\ 1.06 \end{bmatrix}$	$\begin{bmatrix} 2.21 \\ 1.62 \end{bmatrix}$

From the Point of View of ML

In Computer and Graphics Program

We Manipulate the space rotates it in different angle// to represent and calculate the new dimensions



What is Transformation?

Linear transformation
function

5 25

2 $f(x)$ 4

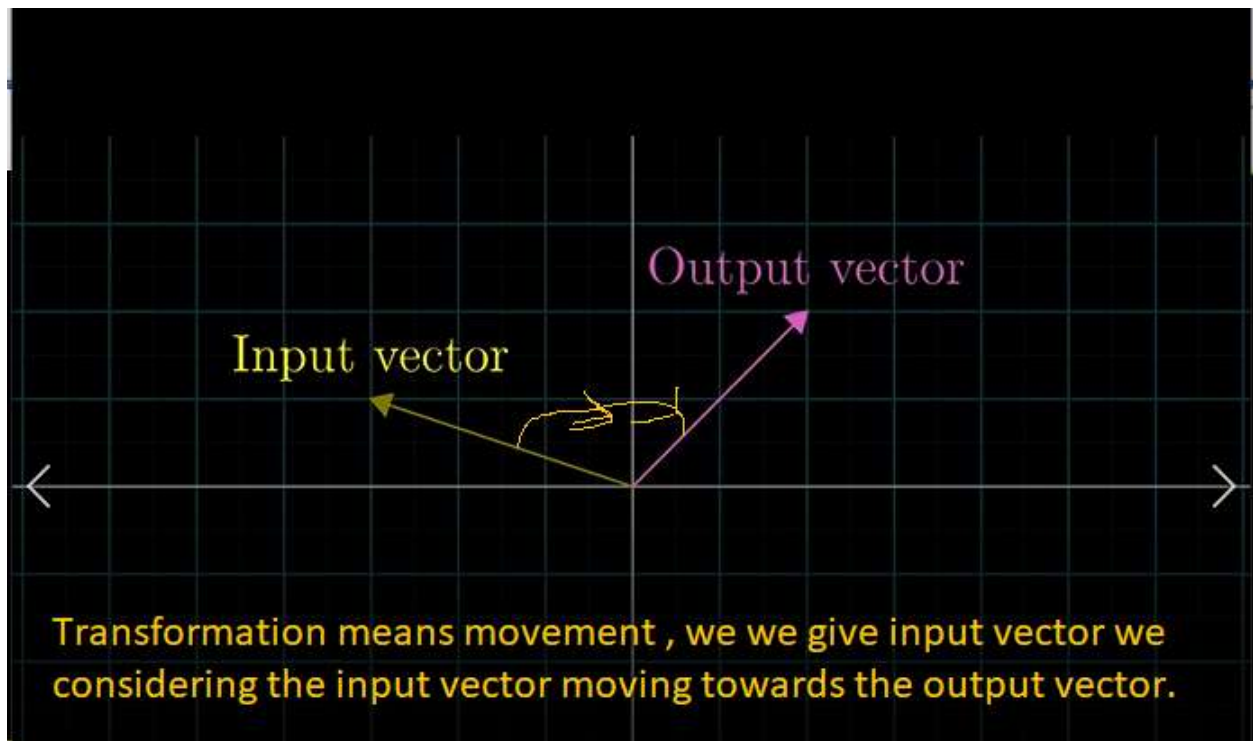
-3 9

It takes the Input and give the Output.

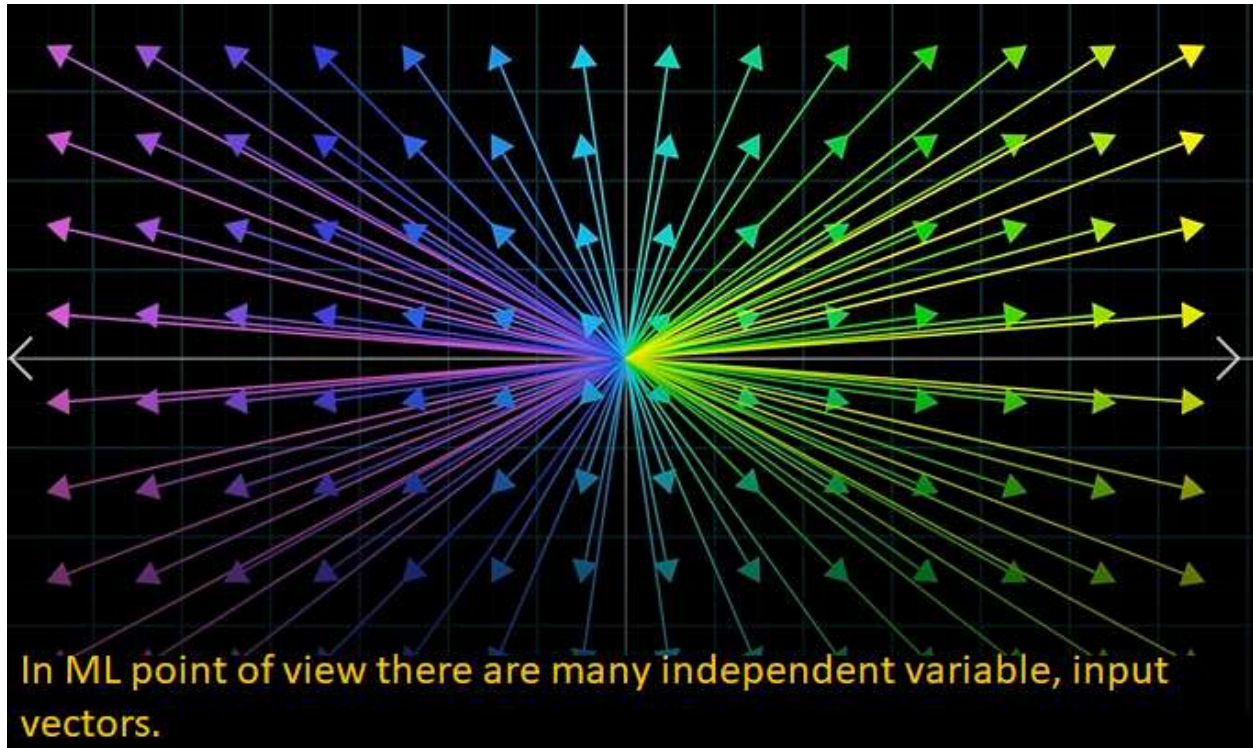
$\begin{bmatrix} 5 \\ 7 \end{bmatrix}$ $L(\vec{v})$ $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$
Vector input Vector output

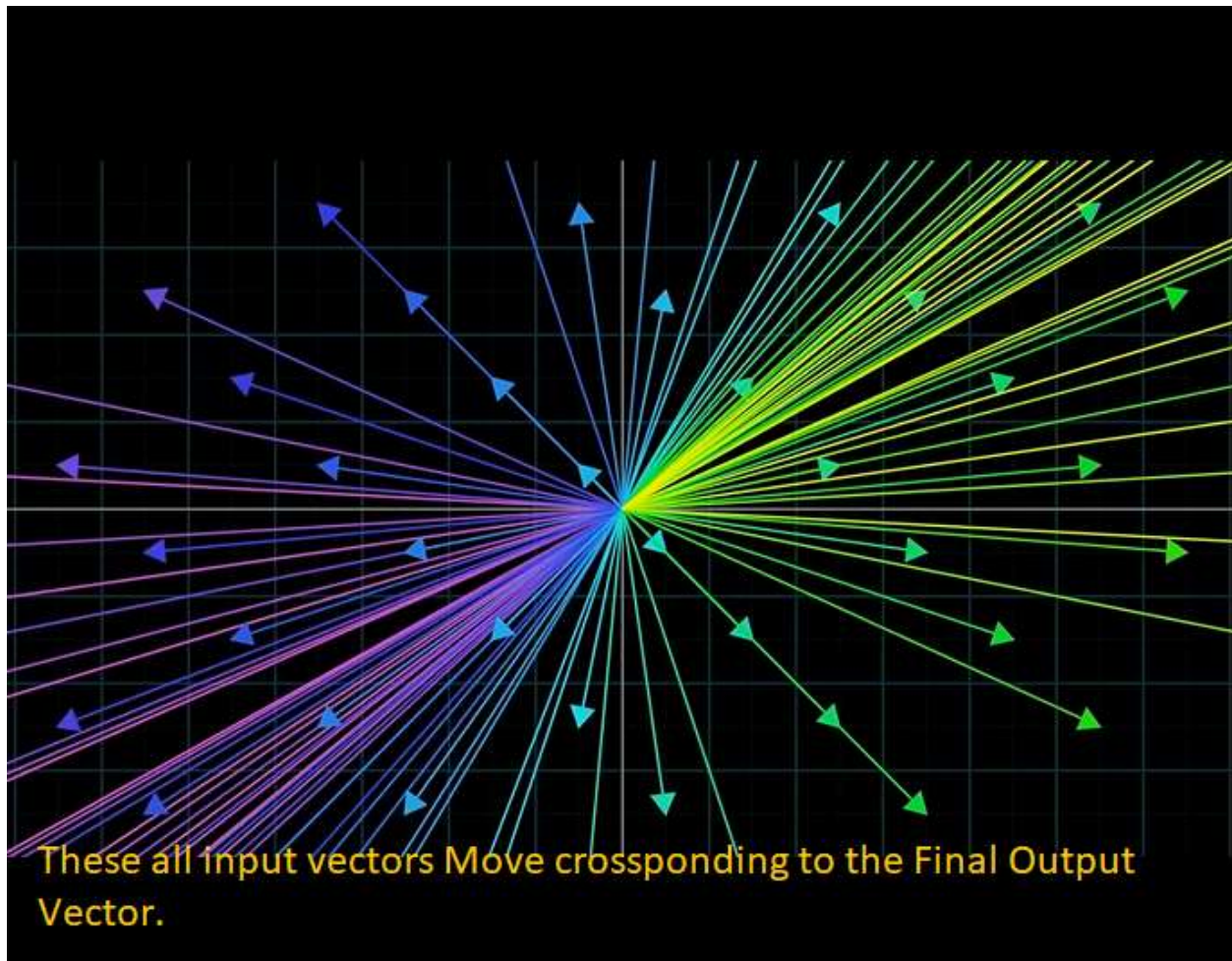
In Linear Algebra It takes the Vector input and Returns the Vector Output.

TRANSFORMATION ---- MOVEMENT –FLOW



Transformation become more complicated in Multi Dimensions





In Machine learning and Deep Learning we will use many a times words.....

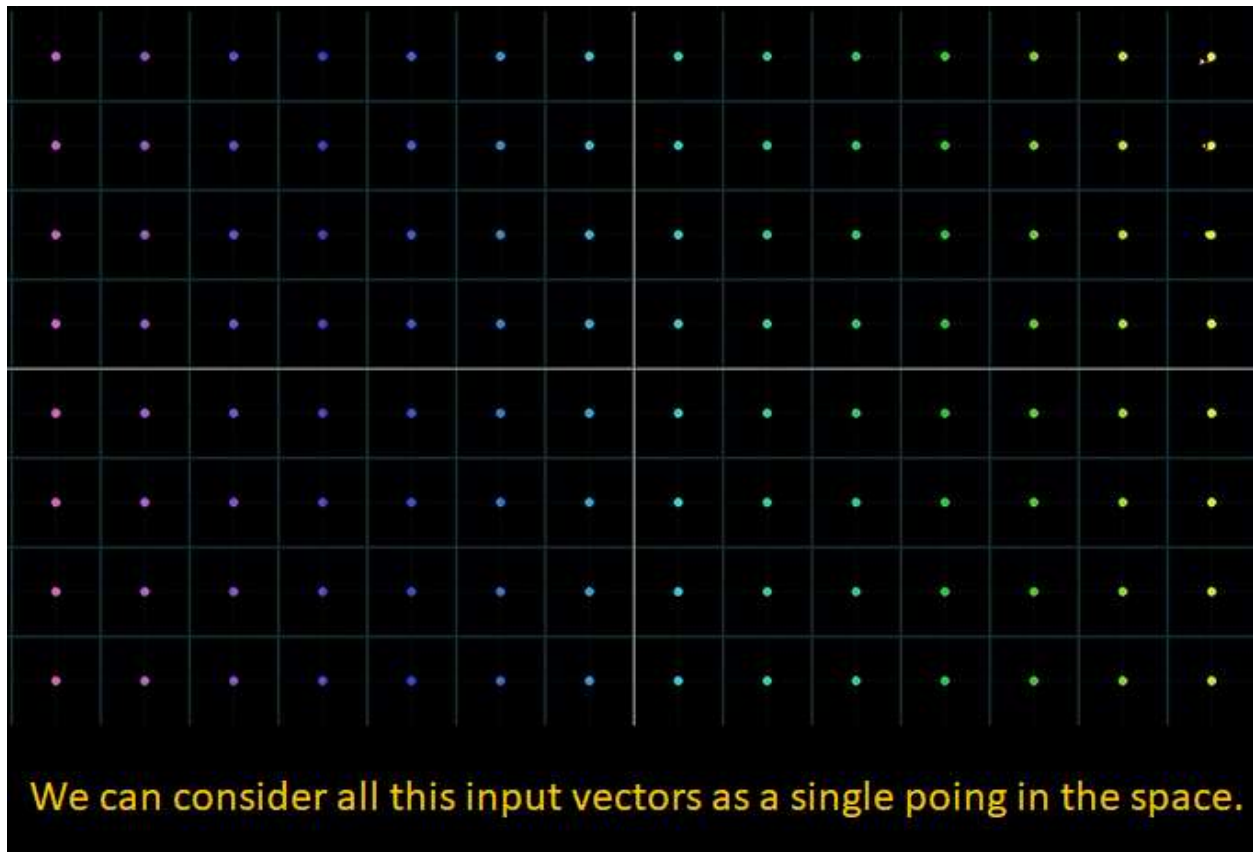
- Transformation
- TENSOR FLOW Networking ...
- FLOW OF Tensor,

Tensor is an object which transform as a project of vectors under rotation.

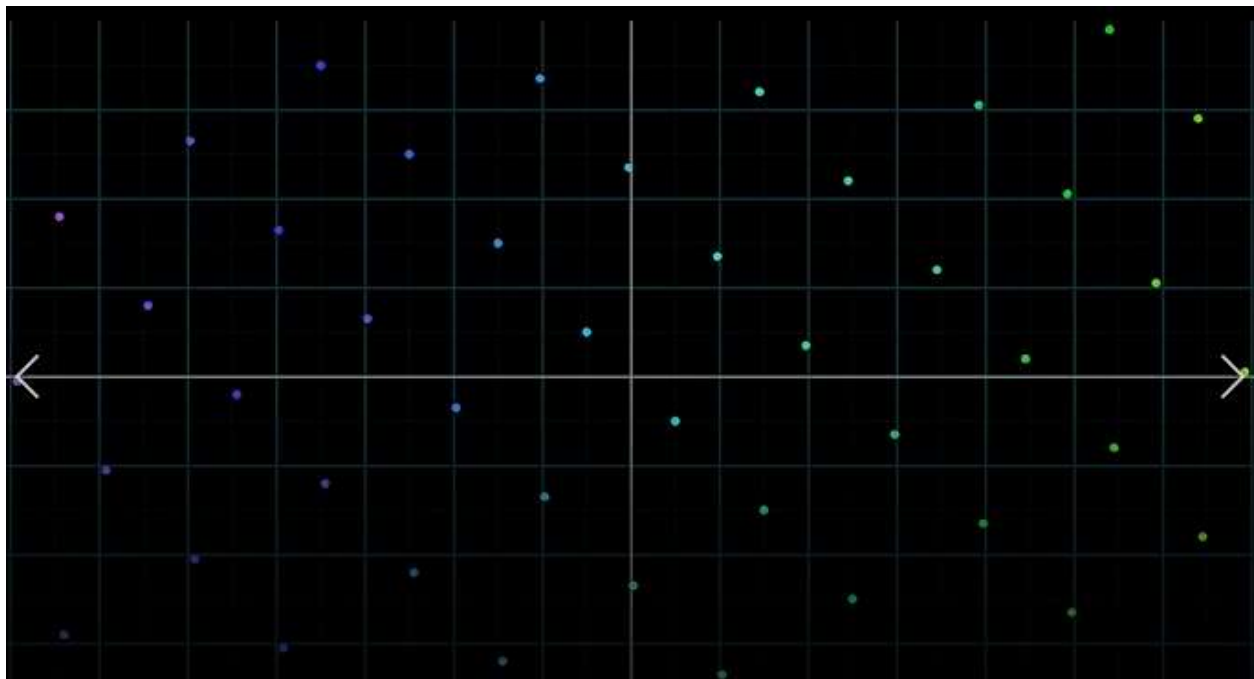
- Input as Tensor means input of VECTOR... MATRICES....

Giving input meansINPUT OF Vector /Matrix.

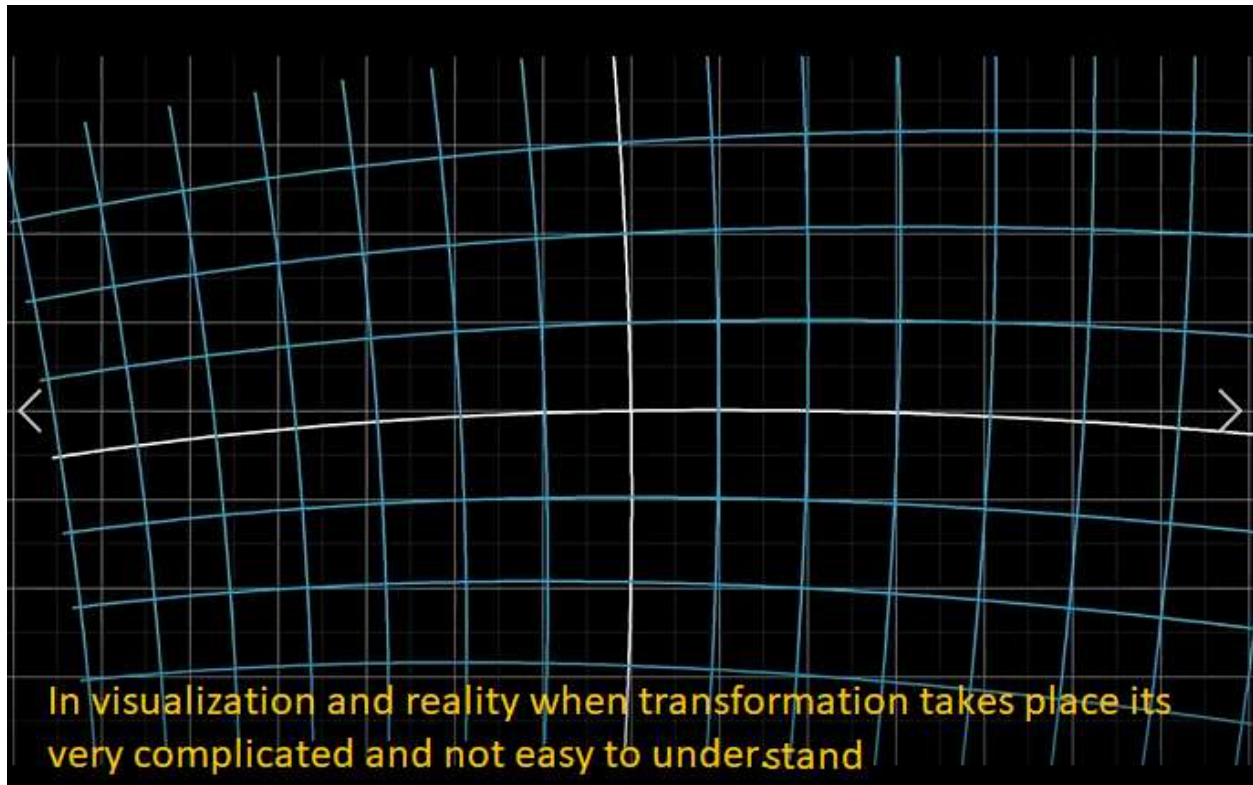
CONSIDERING VETORS AS POINT IN THE SPACE.....



Vectors are in transformation ... means Points are in Transition ..
Points moving one point to some other points.



Transformation in Multi-dimensional space...

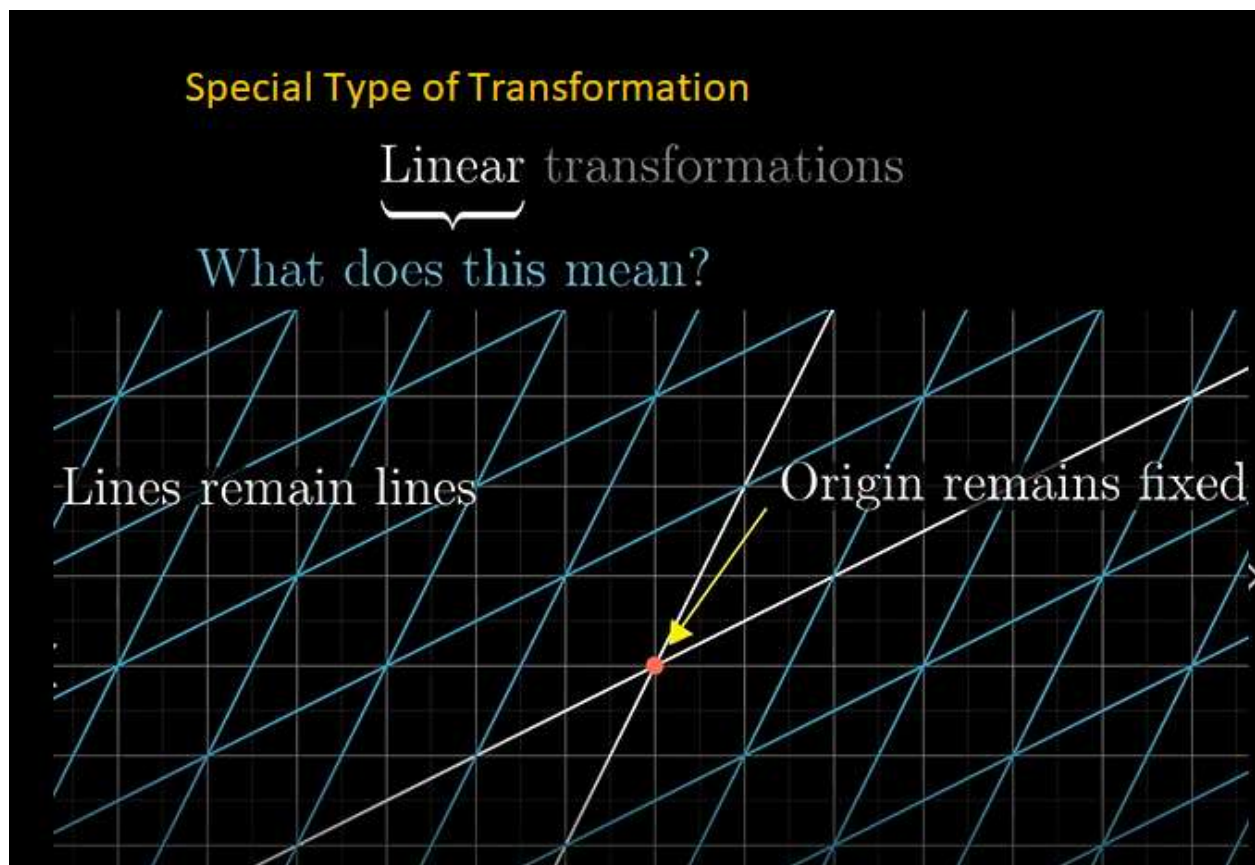


Moving around all the points in the multi-dimensional space.

arbitrary Transformation----(based on random choice)- That is interesting (but difficult to understand) about 4D rotations. Looks very complicated.

To simplify the Mathematical calculation Some assumption has been made. Which bring the concept of Linear Transformation...

LINEAR TRANSFORMATION



FINALLY, GRID LINE REMAIN PARALLEL AND EVENLY SPACED.

Three-dimensional transformation

$$\begin{bmatrix} 2 \\ 6 \\ -1 \end{bmatrix}$$

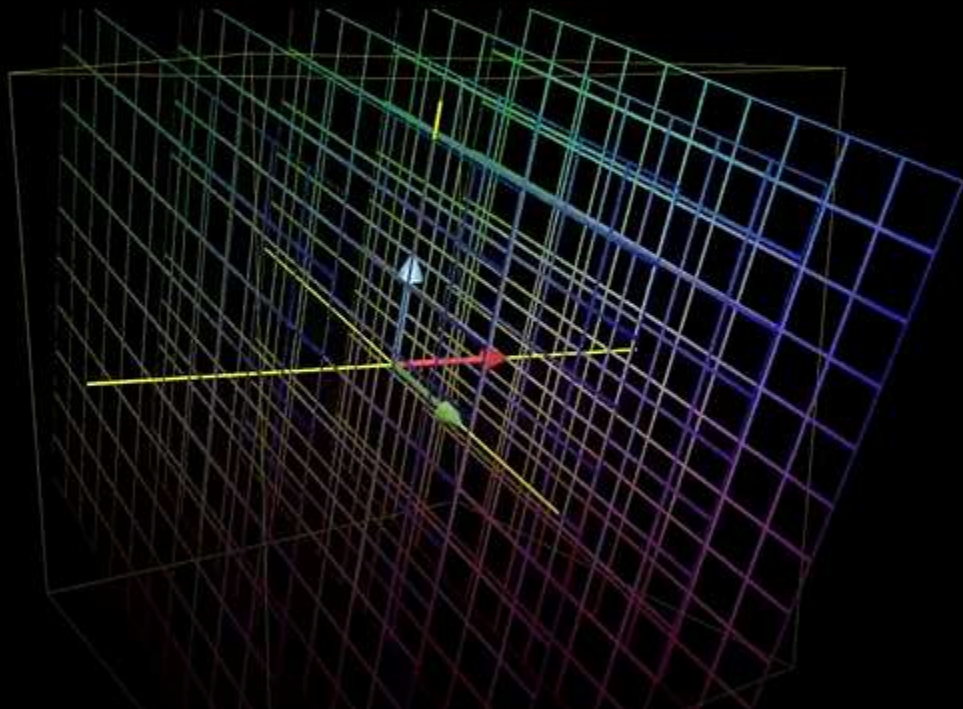
Input

$$L(\vec{v})$$

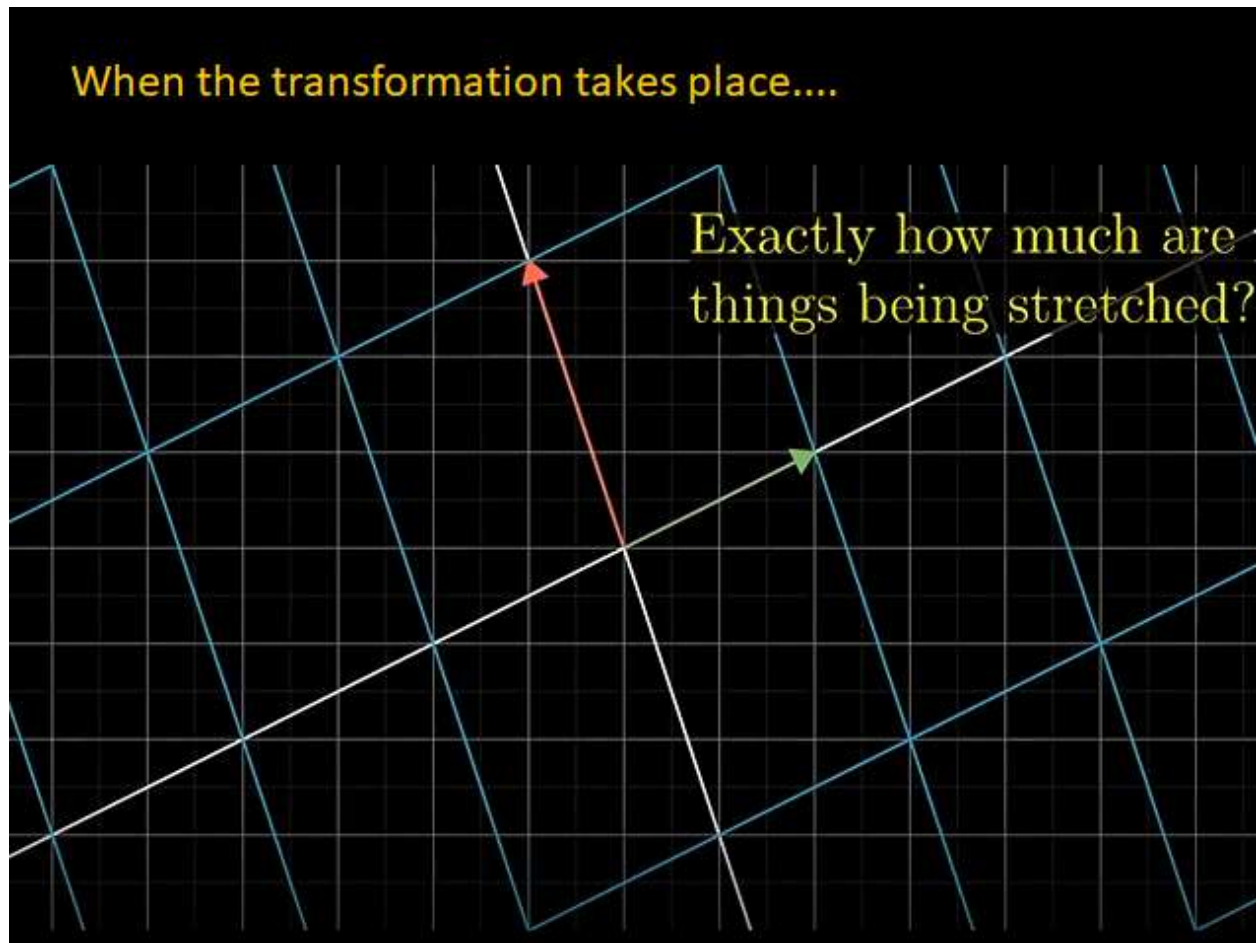
$$\begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$$

Output

3D Transformation



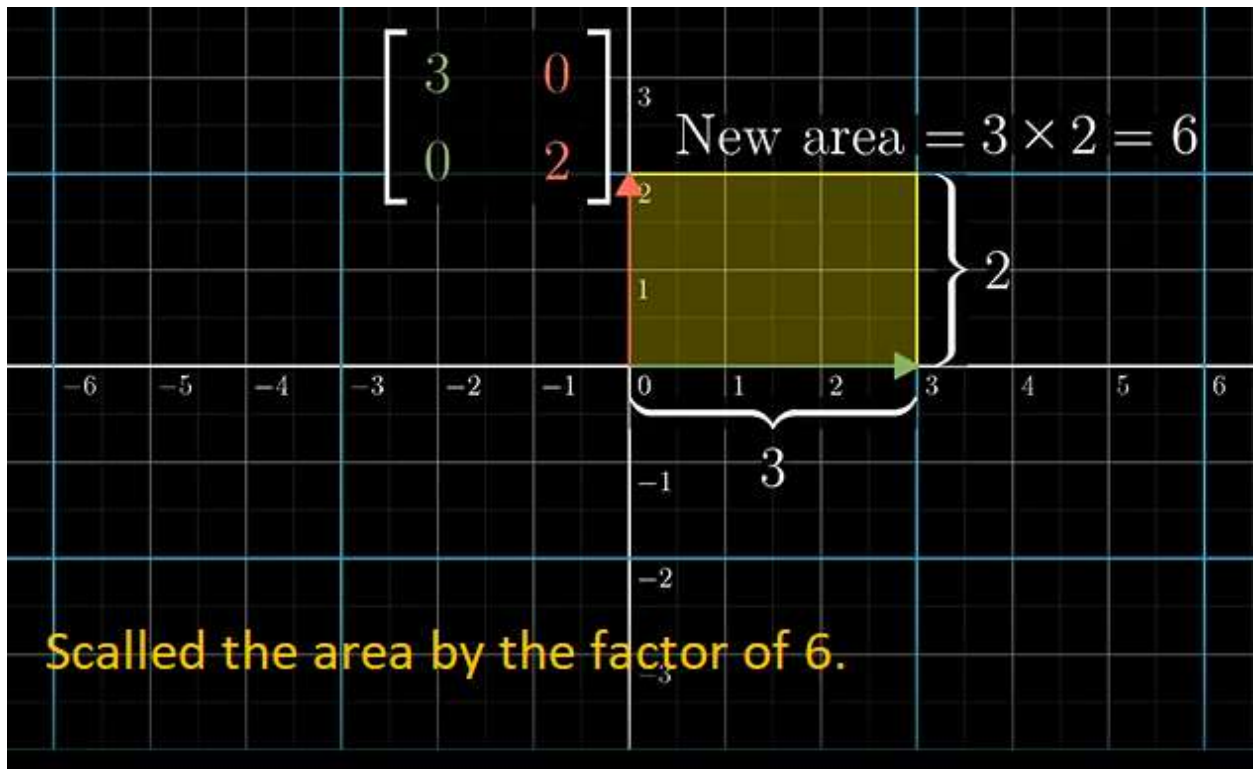
Determinant of Transformation Matrix



Determinant is the Total Area or volume during the transformation.

How much are **areas** scaled?

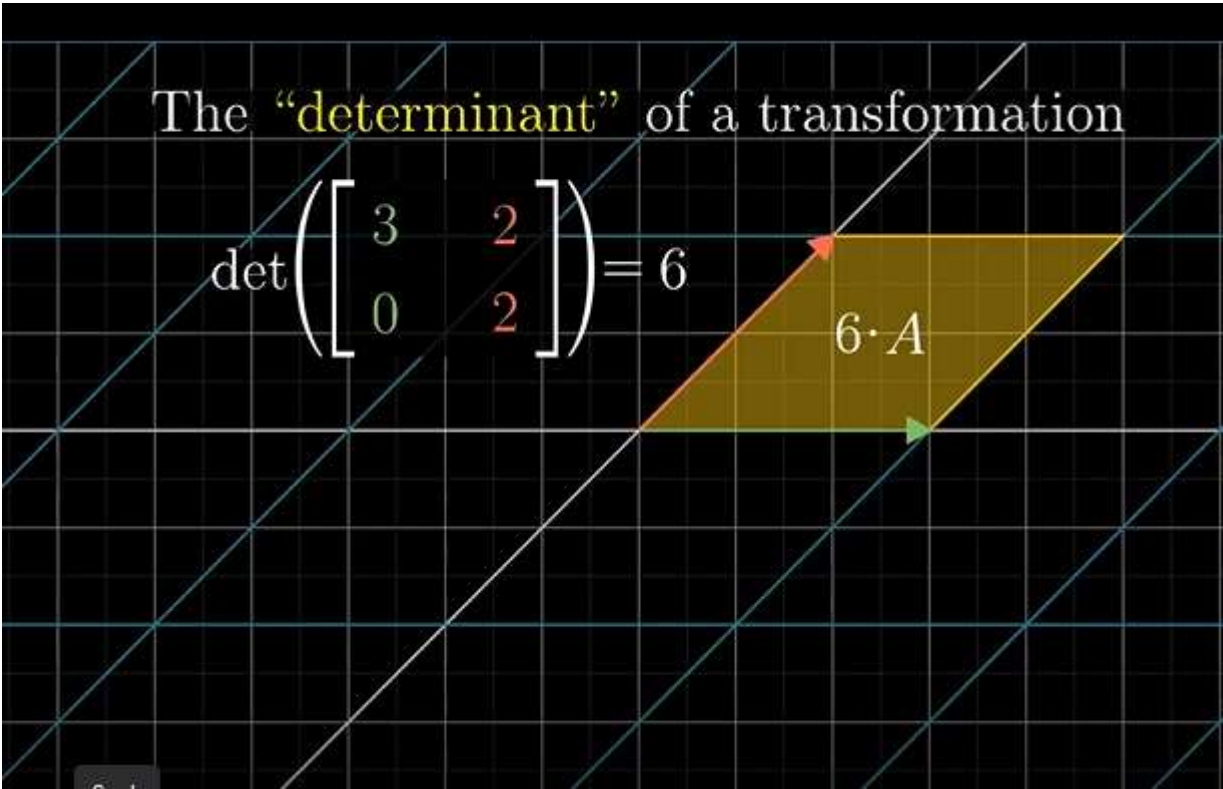
Take the example of this matrix as input variable



The “determinant” of a transformation

$$\det \begin{pmatrix} 3 & 2 \\ 0 & 2 \end{pmatrix} = 6$$

$6 \cdot A$

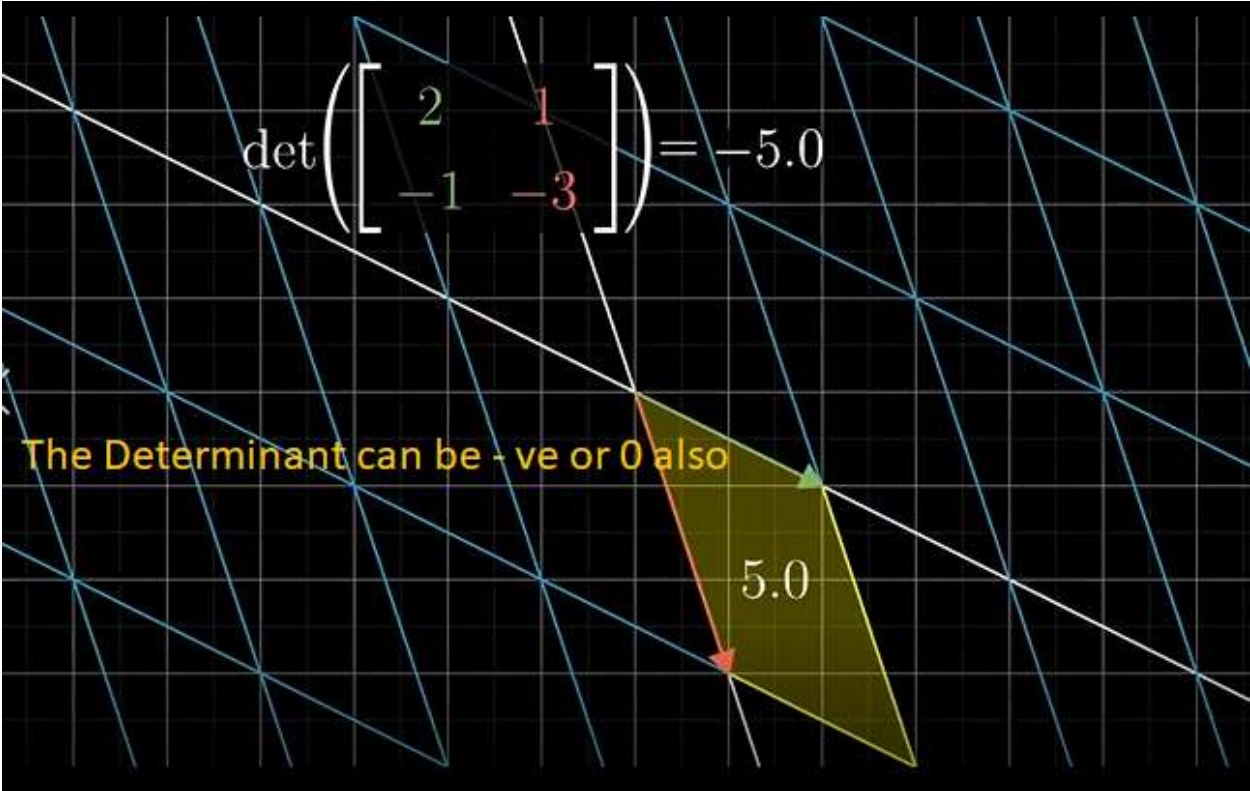


The diagram shows a 2D coordinate system with a grid. A unit square is transformed into a parallelogram. The original square is outlined in white, and the transformed parallelogram is shaded in olive green. The vertices of the parallelogram are at (0,0), (3,0), (5,2), and (3,2). A red arrow points from the origin (0,0) to the top-right vertex (5,2), and a green arrow points from the origin to the top-left vertex (3,2). The area of the parallelogram is labeled as 6 · A.

$$\det \begin{pmatrix} 2 & 1 \\ -1 & -3 \end{pmatrix} = -5.0$$

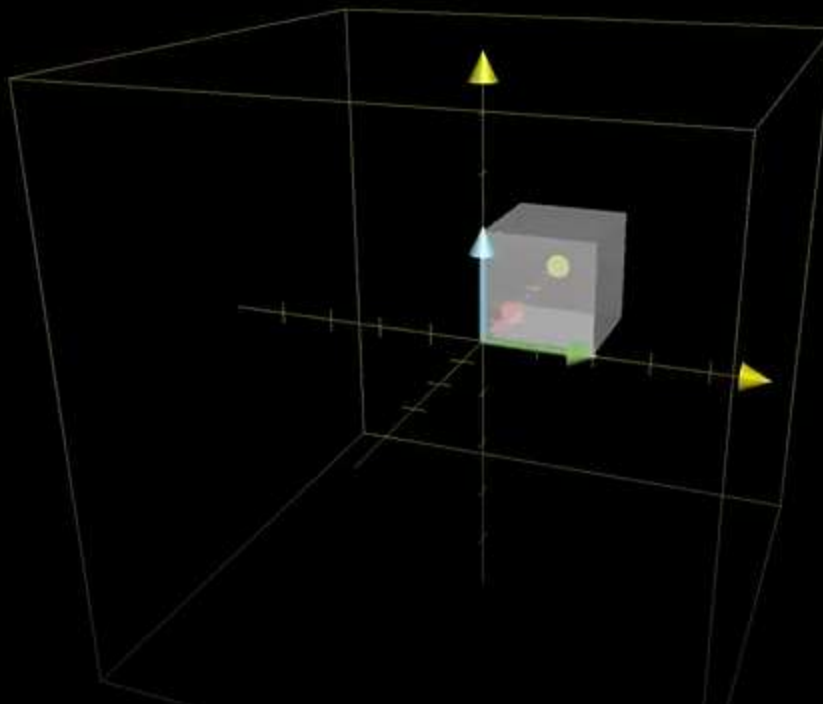
The Determinant can be -ve or 0 also

5.0

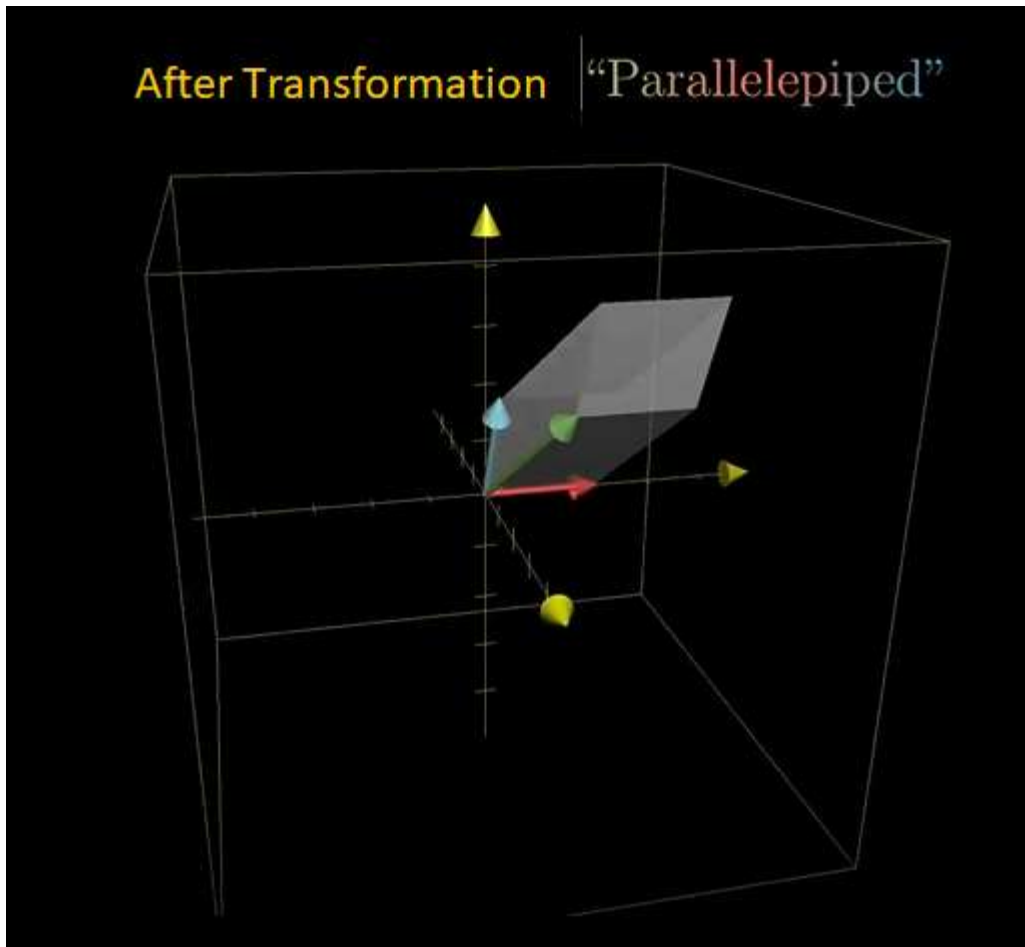


The diagram shows a 2D coordinate system with a grid. A unit square is transformed into a parallelogram. The original square is outlined in white, and the transformed parallelogram is shaded in olive green. The vertices of the parallelogram are at (0,0), (2,1), (-1,-3), and (1,-2). A red arrow points from the origin (0,0) to the top-right vertex (2,1), and a green arrow points from the origin to the top-left vertex (-1,-3). The area of the parallelogram is labeled as 5.0.

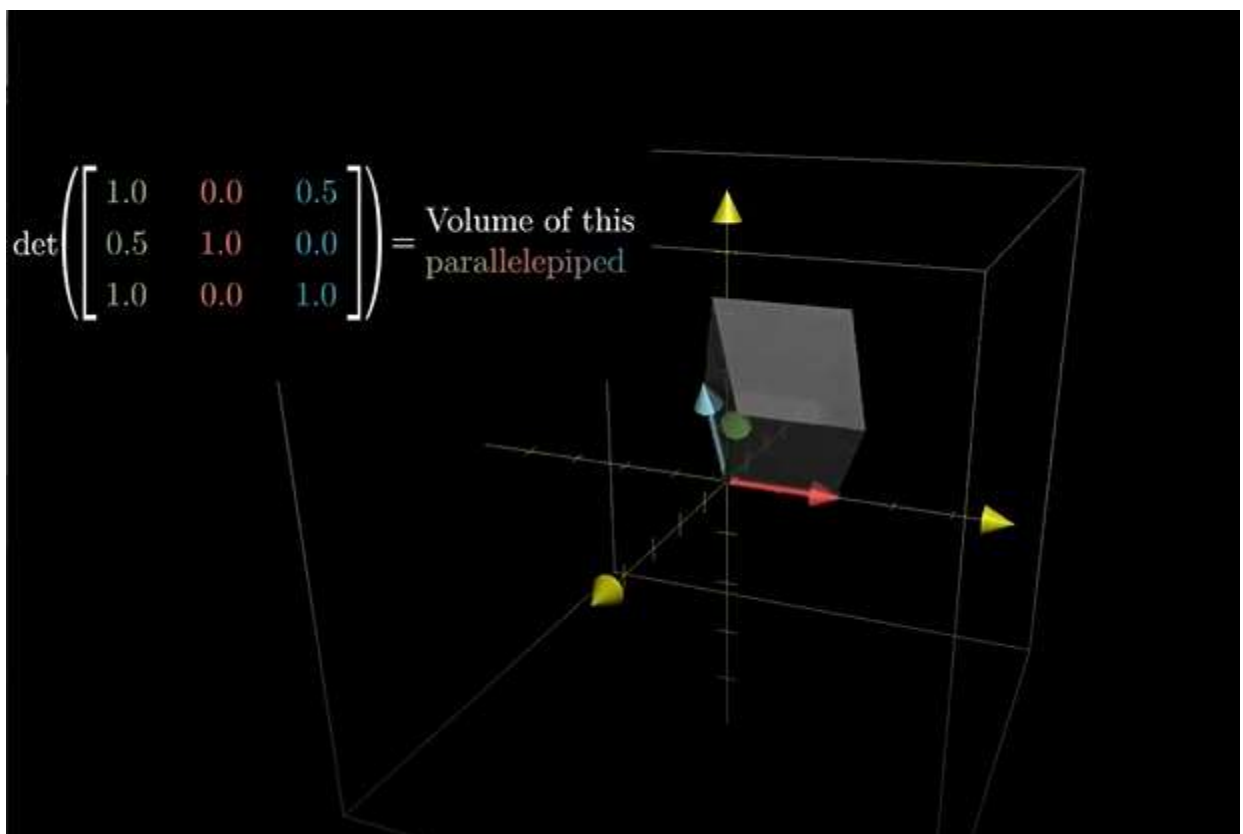
In 2D "Area" Get scaled , in 3D "Volume"
Get Scaled



After Transformation | “Parallelepiped”



$$\det \begin{pmatrix} 1.0 & 0.0 & 0.5 \\ 0.5 & 1.0 & 0.0 \\ 1.0 & 0.0 & 1.0 \end{pmatrix} = \text{Volume of this parallelepiped}$$



Determinant is the volume of parallelepiped

DOT PRODUCTS BETWEEN MATRICES

DOT PRODUCTS

Two vectors of the same dimension

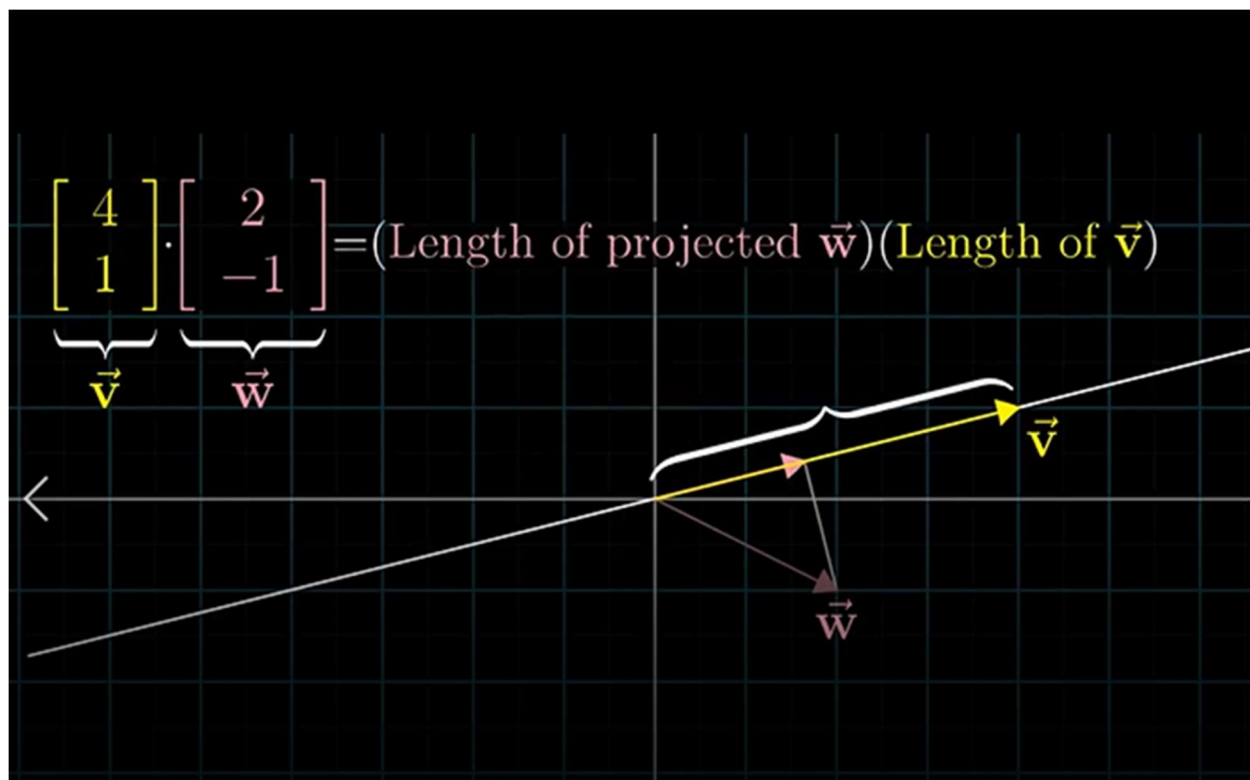
$$\begin{bmatrix} 2 \\ 7 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 8 \\ 2 \\ 8 \end{bmatrix} = 2 \cdot 8 + 7 \cdot 2 + 1 \cdot 8$$

Dot product

Two vectors of the same dimension

$$\begin{bmatrix} 6 \\ 2 \\ 8 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 8 \\ 5 \\ 3 \end{bmatrix} = 6 \cdot 1 + 2 \cdot 8 + 8 \cdot 5 + 3 \cdot 3$$

DOT PRODUCT GRAPHICAL REPRESENTATION



If two vectors are in same direction the dot product is +ve.

if two vectors are in opposite direction the dot product is -ve.

dot product can be 0 also.

dot product application used internally finding the independent variables co-linearity, significance, PCA(principal component analysis

#dot products between two matrices, or two vectors means—how much they the pulling or pushing in some direction.

If Dot product +ve , they are pulling in same direction.

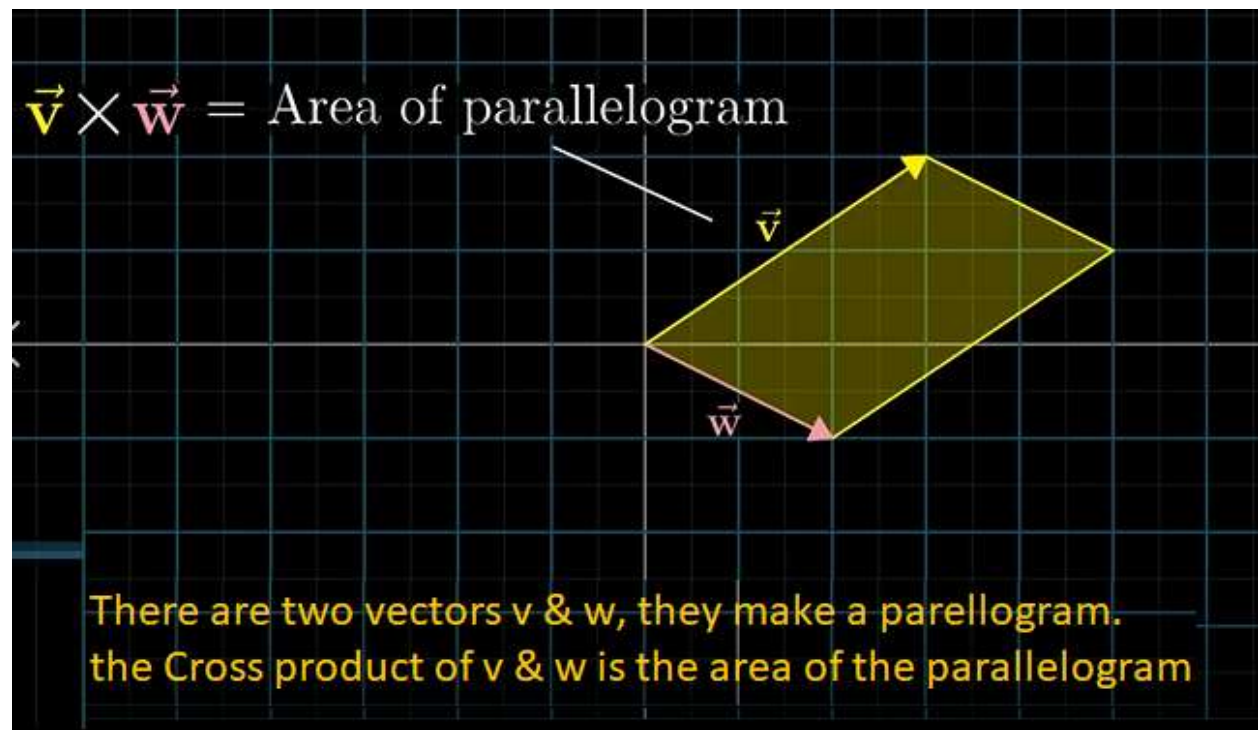
If Dot product -ve , they are pushing in different direction.

If Dot product is +1 they are same value , in same direction, , equal significance.

If Dot product is -1 they are same value , in opposite direction.

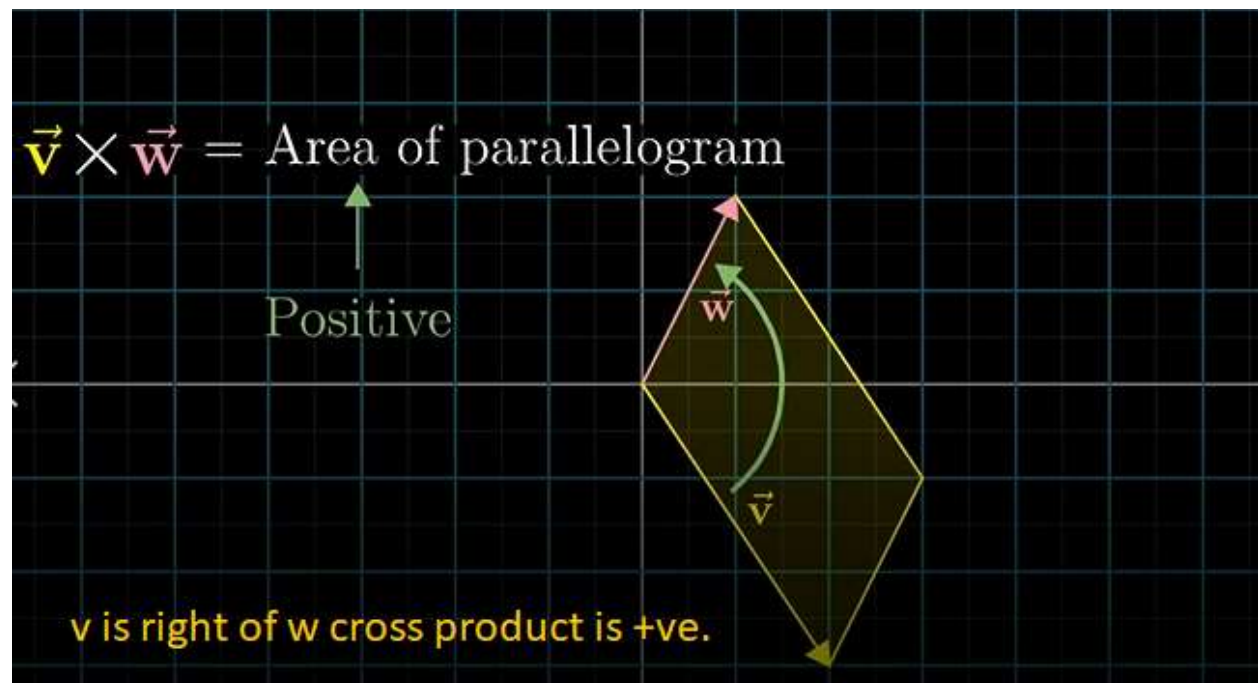
If Dot product is 0 no pulling or pushing , No impact.

CROSS PRODUCT



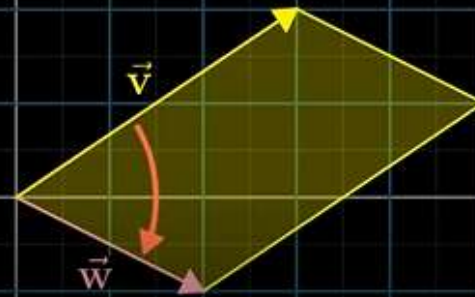
CROSS PRODUCT IS APPLIED FINDING AREA, VOLUME ...

Determinant of Final Matrices... is Area, Volume.



$$\vec{v} \times \vec{w} = - \text{Area of parallelogram}$$

Negative

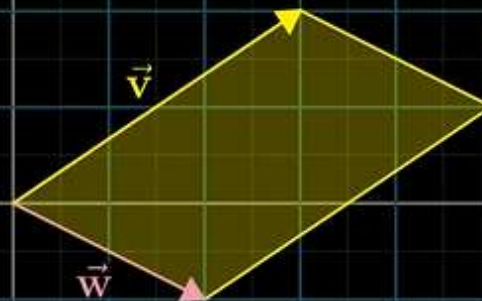


v is on the left of w , the cross product is -ve.

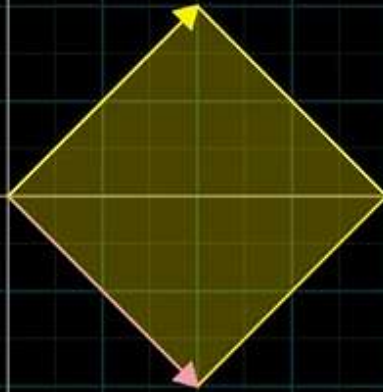
Vector order matters in cross product. (sign get changed + ve to -ve)

$$\vec{v} \times \vec{w} = - \vec{w} \times \vec{v}$$

Order matters

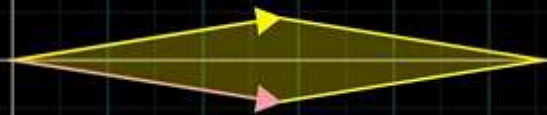


✓ More perpendicular $\Rightarrow \vec{v} \times \vec{w}$ is bigger



help is calculating co-relation coefficient .. here it is high

✓ Similar direction $\Rightarrow \vec{v} \times \vec{w}$ is smaller



Lower corelation coefficient

